

A lightweight and reliable framework toward real-time student engagement predictions in learning analytics

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Abstract

Learning analytics can enable the provision of meaningful feedback based on the collected data, help educators to make decisions with and about learners, and improve learner performance. Student engagement predictions are a key factor in generating feedback for real-time learning analytics applications, such as dashboards. However, most previous work has been based on a heavy deep learning model, which results in challenges for deployment in real-time applications (a resource efficiency requirement in reliable AI). This paper proposes a lightweight deep-learning framework for predicting student engagement in video to address this limitation. The proposed method uses customized MobileNetV2 as the backbone, with an input size of 32 by 32 by 3, to extract features from consecutive video frames. Multi-Scale attention – Residual (MUSER) is used to capture global information and contextual representation of the extracted features. Finally, LSTM examines the temporal variations in video frames and yields the prediction result. We use the DAISEE dataset, the most popular dataset in the learning analytics community, to evaluate the proposed framework. Experimental results demonstrate that the proposed method achieves good accuracy while significantly reducing the model size compared to other approaches.

Keywords: Learning analytics; feedback; lightweight deep learning framework; student engagement predictions; reliable AI

1. Introduction

Comprehensive data are now gathered as students employ digital learning platforms in their courses. This greatly enhances options to comprehend how students learn better [Vytasek et al. \(2020\)](#). The learning analytics community analyses this information to ground recommendations and clarifies learning processes [Roll and Winne \(2015\)](#) for enhanced learning environments [Baker and Inventado \(2014\)](#); [Papamitsiou and Economides \(2014\)](#); [Verbert et al. \(2013\)](#). One major issue in this work is the demand for additional, comprehensive data regarding each student’s learning procedures to create timely and valuable feedback for teachers and students [Winne et al. \(2017\)](#). Student engagement is a vital priority in

this case. Studies consistently indicate a strong correlation between academic success and engagement [Ketonen et al. \(2016\)](#).

Student engagement encloses the learning environment’s emotional, cognitive, and behavioural states [Pilotti et al. \(2017\)](#). Behavioural engagement demands in-class activities, perseverance and emphasizing effort [Okur et al. \(2017\)](#), while cognition implicates education skills such as storage, perception, retrieval, and processing [El Kerdawy \(2020\)](#). Active participation is controlled by affective conditions [Mukhopadhyay et al. \(2020\)](#), whereby negative emotions such as frustration and boredom lead to disengagement, while positive emotions such as interest and happiness improve engagement and focus [Gupta et al. \(2023\)](#). Various approaches have been developed to automate determination of engagement in on-line schooling, classified into computer vision-based and sensor-based methods. Computer vision-based techniques have garnered significant interest, and they have in turn been categorised into video-based and image-based approaches. The image-based methods depend only on spatial data from a single frame or image, which is a notable drawback. Since engagement prediction is spatiotemporal behaviour, video-based strategies have become more widely adopted and efficient for detecting student engagement [Shiri et al. \(2024\)](#).

Video-based approaches generally employ deep learning (DL) and/or machine learning (ML) techniques. ML techniques create features and use handcrafted patterns for student engagement computation [Bosch \(2015\)](#), while DL methods understand features from training data, allowing the process to distinguish variations [Zhang et al. \(2019b\)](#). DL techniques outperform ML in tasks demanding affective condition prediction.

The contributions of this paper are two-fold. Firstly, this study introduces a new lightweight and reliable framework for detecting students’ engagement from video by incorporating the strengths of Customized MobileNetV2 with Multi-Scale attention – Residual (MUSER) attention and LSTM in online learning environments. Secondly, we describe an evaluation of our method using comprehensive experiments on bench-marking datasets, including the DAISEE dataset. The experiment results indicate significant improvements in the number of model sizes with only 0.66 M parameters while keeping a good accuracy of 57.49% compared to the state-of-the-art methods.

2. Related work

Student engagement has been a critical subject in academic research. Computer vision (CV) is presented as a practical method for automated student engagement prediction [Selim et al. \(2022\)](#). At the moment, DL is broadly exploited in CV [Ramamurthy et al. \(2022\)](#) because of its superior performance compared to other techniques [He et al. \(2020\)](#); [Qi et al. \(2021\)](#). These DL approaches are used for engagement detection, providing video data to convolutional neural networks (CNNs) to predict engagement. In [Wu et al. \(2020\)](#), Wu et al. presented a multi-instance learning method in the EmotiW database for engagement detection. Zhu et al. and Huang et al. [Huang et al. \(2019\)](#) introduced the Deep Engagement Recognition Network (DERN), which extracts features from OpenFace [Baltrusaitis et al. \(2016\)](#) and combines them with temporal convolution to determine student engagement on an online education platform, reaching a precision of 60% in classification with four-level of engagement. Gupta et al. [Gupta et al. \(2019\)](#) identified the faces of students with the Max-Margin Face prediction method, and students’ faces were categorised found on

facial expression into four classes, namely, High Negative Affect, High Positive Affect, Low Negative Affect, and Low Positive Affect, employing an improved version of Inception V-3 Model. Zhu et al. [Zhu et al. \(2020\)](#) introduced a Hybrid Attention-based Gated Recurrent Unit (GRU) network to classify features from the EmotiW database’s video and predict the engagement class. Binh et al. [Binh et al. \(2019\)](#) attained 80.8% using a training CNN and transfer learning with VGG19 to classify students’ engagement based on their posture and gestures in the school. Nezami et al. [Nezami et al. \(2019\)](#) trained a simple CNN to recognize engagement in private image sets. Schulc et al. [Schulc et al. \(2019\)](#) created a CNNLSTM method that detected nonattention and attention for commercials.

The majority of the databases employed for evaluating student engagement have either not been publicly available or are small, making comparison with other methods difficult and has restricted the expansion of this field. To address these challenges, Gupta et al. [Gupta et al. \(2016\)](#) presented the Database for Affective States in E-Environments (DAiSEE), which was employed for training numerous DL methods involving the Convolutional 3D (C3D) [Tran et al. \(2015\)](#), InceptionNet [Szegedy et al. \(2015\)](#) and long-term recurrent convolutional networks (LRCN) [Donahue et al. \(2015\)](#), reaching 46.4%, and 57.9% accuracy respectively in four classes of engagement prediction. Moreover, student engagement prediction has evolved as one of the demands in Emotion Prediction in the Wild (EmotiW) [Dhall et al. \(2018\)](#); [Dhall \(2019\)](#) since 2018, promoting this field’s evolution. Authors in [Zhang et al. \(2019a\)](#) enhanced the Inflated 3D Convolutional Network (I3D) [Carreira and Zisserman \(2017\)](#) and achieved greatest precision in the two-level engagement category, namely not engaged or engaged. Authors in [Niu et al. \(2018\)](#) developed a new Gaze-AU-Pose (GAP) unit and employed a Gated Recurrent Unit (GRU) [Cho \(2014\)](#) layer and recurrent neural network (RNN) to predict engagement. Liao et al. [Liao et al. \(2021\)](#) presented Deep Facial Spatio-Temporal Network (DFSTN) to detect the engagement class during online education, employing the SE-ResNet-50 model for spatial features retrieval and the LSTM along with global attention for temporal data retrieval. The DFSTN approach achieved a precision of 58.84% in four engagement classes on the DAiSEE database. Geng et al. [Geng et al. \(2019\)](#) proposed a Convolutional 3D (C3D) model with Focal Loss [Lin et al. \(2017\)](#) to predict engagement from videos.

Although the above methods perform excellently on the DAiSEE dataset, none is designed for mobile applications because they rely on the heavy deep learning framework. We propose a new lightweight deep learning method to address this gap. Lightweight deep learning models demand fewer computational resources, providing efficient power consumption, allowing them to operate smoothly on mobile devices with limited hardware resources, and advancing reliable AI by delivering consistent performance on low hardware resources (resource efficiency). Furthermore, lightweight deep learning models are available across multiple users and devices, ensuring the AI system can accommodate many users simultaneously without performance degradation, enhancing overall reliability (scalability of reliable AI).

3. Methodology

Figure 1 presents the proposed model’s structure. Frames are taken from each video and transformed into spatial features by improved MobilNetV2. A unique instance of the im-

proved MobilNetV2 model is applied to every frame. Next, MUSER is utilized to learn the extracted features' global information and contextual representation. Next, LSTM is used to understand the temporal information. After finishing the training prsocess, the proposed method recognizes the correlation spatiotemporal features with engagement levels. The details of each component framework are discussed in the following subsection.

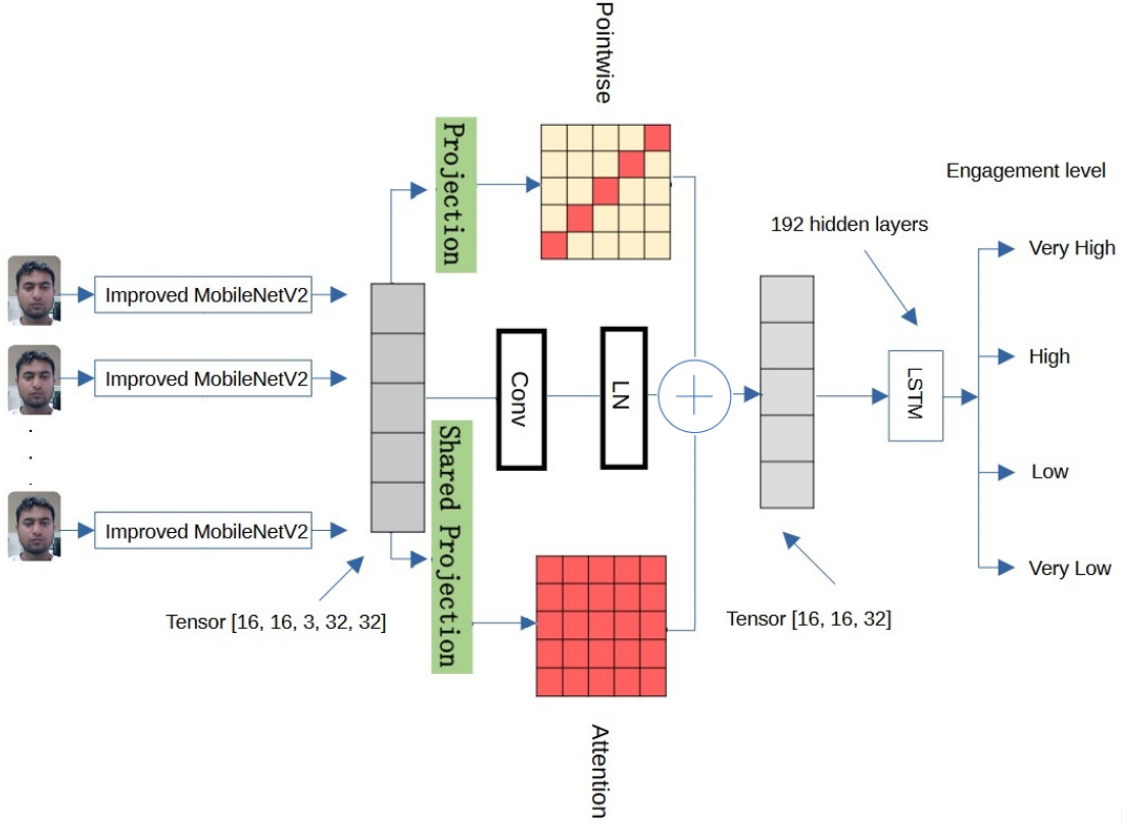


Figure 1: The proposed framework.

3.1. Customized MobileNetV2

Along with the fast development of CNN, the model structure is complicated and requires a lot of calculations. The need for hardware resources is more and more. The training process cannot be developed on resource-constrained systems, so the recent research approach is high-speed and lightweight [Zhu et al. \(2019\)](#). One approach is compressing the trained network and getting a corresponding lightweight network. Another strategy is to create a small network for training, and the MobileNet V2 is a suggestive small model. MobileNet V2 compensates accuracy and model size compared to MobileNet V1 and MobileNet V3. MobileNetV2 presents the linear bottleneck and residual structure in the network.

The standard residual architecture lowers the dimension first and then expands it. The residual structure in MobilenetV2 uses the contrary order. First, a 1×1 convolution layer

is employed to increase the dimension. Then, a 3×3 deep convolution layer is employed to substitute 3×3 convolution, which can significantly improve the effectiveness and decrease the computation of the model. Lastly, a 1×1 convolution layer is employed to decrease the dimension. In MobilenetV2, the ReLU is removed in the final layer and is replaced with the linear activation function. This approach can handle the issue of data loss. Moreover, extension coefficients are presented to regulate the model size [Zhang et al. \(2022\)](#). When the stride is 2, no shortcut is used. In the proposed MobileNetV2 network, the shortcut is added with a 1×1 convolution layer and batch normalization layer in the bottleneck block with a stride of 2 (see Figure 2). The extra residual branch allows further improvement in the network performance.

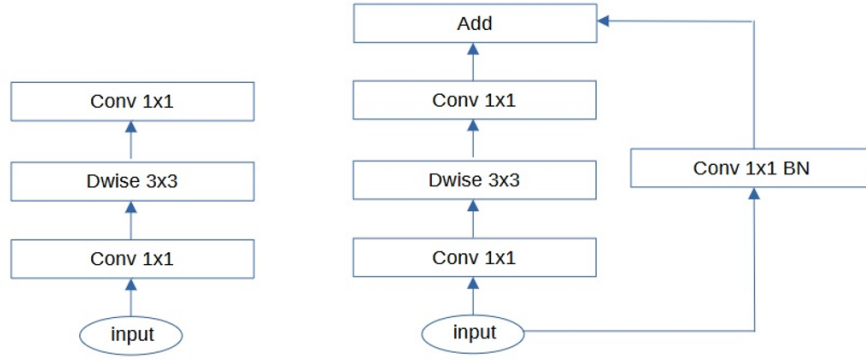


Figure 2: The original bottleneck (left) and the proposed bottleneck (right).

In the original MobileNetV2, the width multiplier α is a hyper-parameter that changes the number of filters in depthwise convolutions. This multiplier affects all layers except the final convolutional layers. When $\alpha = 1.0$, the network utilizes the original number of filters. When α is less than 1.0, the number of filters in each convolutional layer is scaled down by the factor α . In TensorFlow, pre-trained networks weights are available for $\alpha = 0.35$, $\alpha = 0.5$ and $\alpha = 1.0$ [Narduzzi et al. \(2022\)](#). In the proposed MobileNetV2 network, the value of α is 0.35.

3.2. Multi-Scale attention – Residual (MUSER)

The self-attention mechanism is extremely efficient in sequence-to-sequence learning and significantly enhances many tasks. However, it has its own drawbacks. The attention in deep layers is likely to focus in a single instance, yielding to the problem of expressing long sequences.

Multi-scale attention (MUSE) interprets parallel multi-scale expressing learning on sequence information. The original MUSE has two variants: MUSE-simple and MUSE. MUSE-simple includes the essential concept of a parallel multi-scale sequence describing data, and it encrypts the sequence in parallel, regarding various scales, supported by point-wise transformation and self-attention mechanism. MUSE is made from MUSE-simple and

interprets a combination of the self-attention mechanism and convolution for learning sequences describing from various scales.

MUSE concentrates on machine translation and has substantially improved performance over Transformer, particularly on long sequences. More significantly, MUSE’s success in the application needs multi-scale attention and complex considerations to unify semantic space.

Under a standard setting, the proposed MUSE model surpasses all earlier methods on three principal machine translation tasks. Moreover, MUSE can accelerate inference due to its parallelism [Zhao et al. \(2019\)](#).

In this work, we extend the MUSE simple with the layer normalization and residual mechanism to create a new attention mechanism, MUSER. The MUSER accepts the output of improved MobileNetV2 as input and creates the output data in a fusion form:

$$Y = F(X) \oplus \text{Attention}(X) \oplus \text{Pointwise}(X) \quad (1)$$

Where “Attention” indicates self-attention, “F(X)” indicates the residual network, and “Pointwise” indicates a position-wise feed-forward network. The residual network consists of two layers: conv 1x1 and Layer Normalization. As mentioned in the previous section, conv 1x1 changes the channels’ number in the feature map. Layer Normalization, a simple normalization technique to enhance the training speed for different neural network models, is used over a mini batch of the output from the conv 1x1 layer. Layer Normalization executes the operation as defined in [pyt; Lei Ba et al. \(2016\)](#):

$$s = \frac{t - E[t]}{\sqrt{\text{Var}[t] + \epsilon}} * \gamma + \beta \quad (2)$$

The standard deviation and mean are computed over the final D dimensions, where D is the dimension of the normalized shape. For instance, if the standardized form is (3, 5), the standard deviation and mean are calculated over the last two dimensions of the input (i.e. input mean ((-2, -1))). γ and β are learnable affine transform parameters of the standardized form if the elementwise affine is True. The typical deviation is computed by the biased estimator.

3.3. Long short-term memory (LSTM)

LSTM presented as a seminal in [Hochreiter and Schmidhuber \(1997\)](#), is a refined version of the RNN specifically designed to address the typical problem of long-term dependency [Shiri et al. \(2023\)](#). LSTMs have proven effective in maintaining data over expanded sequences, thereby resolving the vanishing gradient issue [Barot and Kapadia \(2022\)](#). LSTM processes the result from the prior “time step” along with the current input at a provided “time step” and passes it to the next “time step”. Normally, the final step of the last hidden layer is used for recognition [Minaee et al. \(2019\)](#).

The LSTM framework contains a hidden state described by h, a memory unit symbolised as c, and three different gates: The forget, the input, and the output gate. These gates are essential in managing the input’s information flow, thereby controlling the write and read operations within the LSTM framework. The input gate decides how the internal state is updated depending on the current input and the prior internal state. In contrast, the forget gate determines how much of the previous internal state is retained [Shiri et al. \(2024\)](#).

Lastly, the output gate modulates the effect of the internal state on the overall system [Fang et al. \(2021\)](#). At every time step t , the LSTM obtains an input along with the prior hidden state. Later, it computes activations for the gates and updates both the hidden state and the memory unit.

The improved MobileNetV2 creates spatial features from single frames, the MUSER improves the learning feature, and the LSTM catches the temporal variations in the sequence of frames and creates the detected engagement class. For a provided video sequence, the input to the proposed framework is an $L \times C \times H \times W$ tensor, where C , L , W , and H correspond to the number of channels, number of frames, frame width, and frame height, respectively. The improved MobileNetV2 is the feature extractor from individual frames of the input video. Then, the MUSER enhances the extracted feature from the output of the improved MobileNetV2. Next, the enhanced feature vectors from the consecutive frames are considered as the multi-dimensional input to the consecutive time steps of an LSTM to catch temporal information in the video. The output of the final time step of the LSTM is fed to a fully connected layer to predict the engagement class from the input video.

MUSER improves the performance of engagement prediction for deep learning networks compared to other attention mechanisms such as Squeeze-and-Excitation Networks (SENet) [Hu et al. \(2018\)](#) or Linformer [Wang et al. \(2020\)](#). SENet significantly improves accuracy for existing deep learning networks at a small extra computational cost. Nevertheless, it was designed for image classification and unsuitable for video prediction tasks. Linformer, on the other hand, is created to manage long sequences efficiently, making it appropriate for sequential data like video frames. However, one drawback of Linformer is that it processes sequences in nonparallel. In contrast, the MUSER attention learns sequences in parallel, so it easily captures the change in student behavior in the video data. Also, MUSER uses residual branch, helps mitigate the vanishing gradient problem, and improves features learned [He et al. \(2016\)](#), leading to better performance than the original MUSE.

4. Experiments

The DAiSEE database, presented in [Gupta et al. \(2016\)](#), is employed to measure the proposed method’s performance compared to the prior approaches. The database has 9,068 videos taken from 112 students to identify their states of confusion, boredom, frustration, and engagement. We just concentrate on engagement, with four engagement stages: stage 0 (very low), 1 (low), 2 (high), and 3 (very high). These states were provided by Whitehill et al. [Whitehill et al. \(2014\)](#). The frame rate, length, and resolution of the videos are 30 fps, 10 seconds, and 640×480 pixels. Table 1 displays the allocation of samples in the training, validation, and testing sets as defined in [Gupta et al. \(2016\)](#). Our numerical section uses these sets to fairly compare the proposed framework with prior techniques on the DAiSEE database. We merged 1429 and 5358 videos in the validation and training sets to train the proposed framework model and make predictions on 1784 testing videos [Abedi and Khan \(2021\)](#). The database is highly imbalanced; only 0.63%, 1.61%, and 0.22% per cent of training, validation, and testing sets are in level 0. The extracted frames from the video were downsampled to get $3 \times 32 \times 32$ images. The Adam is applied for parameter optimization with a batch size and learning rate of 16 and 0.001, respectively. The improved MobileNetV2 extracts feature vectors from the continuous frames and sends them to the

MUSER, then LSTM. We executed the experiments using PyTorch [Paszke et al. \(2019\)](#) on a PC with 32 GB memory and an QUADRO M4000 8GB GPU.

Table 1: The DAiSEE database allocation.

	Level 0 Very Low	Level 1 Low	Level 2 High	Level 3 Very High
Total	61	455	4419	3886
Train	46	300	2846	2468
Validate	11	74	712	641
Test	4	81	861	777

The proposed framework’s performance and the comparison results with various methods on the DAiSEE database are shown in Table 2. Some of the prior methods on the DAiSEE database, not evaluated with the four-class engagement, showed accurate metrics [Gupta et al. \(2016\)](#); [Liao et al. \(2021\)](#); [Huang et al. \(2019\)](#); [Wu et al. \(2020\)](#); [Geng et al. \(2019\)](#). In [Gupta et al. \(2016\)](#), the authors provided five models: frame-level InceptionNet, video-level InceptionNet, C3D fine-tuning, C3D feature extraction, and LRCN. The top precision in [Gupta et al. \(2016\)](#) was 57.9% with LRCN, outperforming other competitive networks like ResNet TCN with weighted loss and sampling [Paszke et al. \(2019\)](#).

Table 2: The comparison between various methods on the DAiSEE database.

Model	Accuracy	Parameters
frame-level InceptionNet Gupta et al. (2016)	47.10%	6.79M
video-level InceptionNet Gupta et al. (2016)	46.40%	6.79M
ResNet+TCN with weighted loss Abedi and Khan (2021)	53.7%	11M
LRCN (CaffeNet) Gupta et al. (2016)	57.90%	61M
DFSTN Liao et al. (2021)	58.84%	25.6M
EfficientNet B7 + Bi-LSTM Selim et al. (2022)	66.39%	66M
EfficientNet B7 + TCN Selim et al. (2022)	64.67%	66M
EfficientNet B7 + LSTM Selim et al. (2022)	67.48%	66M
The proposed method (Improved MobileNetV2+MUSER+LSTM)	57.49%	0.67M

The DFSTN [Liao et al. \(2021\)](#) reached a precision of 58.84% on the DAiSEE database, which surpassed LRCN by 0.94%. Abedi and Khan [Abedi and Khan \(2021\)](#) presented ResNet TCN with weighted loss and sampling, which achieved an accuracy of 53.7%. Finally, the highest results attained by using the three models: EfficientNet B7 + LSTM [Selim et al. \(2022\)](#), EfficientNet B7 + TCN [Selim et al. \(2022\)](#), and EfficientNet B7 + Bidirectional LSTM [Selim et al. \(2022\)](#) network on the DAiSEE database were 67.48%, 64.67%, and 66.39%, respectively. As seen in Table 2, the proposed method achieved good accuracy while significantly reducing the model size compared to other approaches. For example, LRCN exceeded the proposed method by only 0.5% but increased the model size almost 100 times.

5. Conclusion

In this paper, we have introduced a novel architecture, Improved MobileNetV2+ MUSER attention + LSTM, for determining engagement levels in online education. We measured the proposed framework's performance and compared it to several state-of-the-art approaches. The proposed architecture significantly reduces the parameters while maintaining good accuracy compared to state-of-the-art approaches on the DAiSEE database.

Student engagement is crucial for tracking the student learning process. The proposed method is lightweight and suitable for mobile applications. Future research is required to integrate the proposed method into the real-time learning applications.

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References

- Pytorch layernorm. <https://pytorch.org/docs/stable/generated/torch.nn.LayerNorm.html>.
- A. Abedi and S. Khan. Improving state-of-the-art in detecting student engagement with resnet and tcn hybrid network. In *Proceedings of the 18th Conference on Robots and Vision (CRV)*, pages 151–157, 2021.
- R. S. Baker and P. S. Inventado. Educational data mining and learning analytics. In *Learning Analytics*, pages 61–75. Springer, Berlin, 2014.
- T. Baltrusaitis, P. Robinson, and L. Morency. Openface: An open source facial behavior analysis toolkit. In *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, pages 1–10, 2016.
- V. Barot and V. Kapadia. Long short term memory neural network-based model construction and finetuning for air quality parameters prediction. *Cybernetics and Information Technologies*, 22(1):171–189, 2022.
- H. Binh, N. Trung, H. Nguyen, and B. Duy. Detecting student engagement in classrooms for intelligent tutoring systems. In *Proc. 23rd Int. Comput. Sci. Eng. Conf. (ICSEC)*, pages 145–149, 2019.
- N. Bosch. Automatic detection of learning-centered affective states in the wild. In *Proceedings of the 20th International Conference on Intelligent User Interfaces (IUI)*, pages 379–388, 2015.

- 294 J. Carreira and A. Zisserman. Quo vadis, action recognition? a new model and the ki-
 295 netics dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern*
 296 *Recognition (CVPR)*, pages 6299–6308, 2017.
- 297 K. Cho. Learning phrase representations using rnn encoder-decoder for statistical machine
 298 translation. *arXiv preprint arXiv:1406.1078*, 2014.
- 299 A. Dhall. Emotiw 2019: Automatic emotion, engagement and cohesion prediction tasks. In
 300 *Proc. 2019 International Conference on Multimodal Interaction (ICMI)*, pages 546–550,
 301 2019.
- 302 A. Dhall, A. Kaur, R. Goecke, and T. Gedeon. Emotiw 2018: Audio-video, student engage-
 303 ment and group-level affect prediction. In *Proc. 20th ACM International Conference on*
 304 *Multimodal Interaction (ICMI)*, pages 653–656, 2018.
- 305 J. Donahue, H. L. Anne, S. Guadarrama, M. Rohrbach, S. Venugopalan, K. T. Saenko,
 306 and T. Darrell. Long-term recurrent convolutional networks for visual recognition and
 307 description. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pages 2625–
 308 2634, 2015.
- 309 M. El Kerdawy. The automatic detection of cognition using eeg and facial expressions.
 310 *Sensors*, 20(12):3516, 2020.
- 311 W. Fang, Y. Chen, and Q. Xue. Survey on research of rnn-based spatio-temporal sequence
 312 prediction algorithms. *Journal of Big Data*, 3(3):97–110, 2021.
- 313 L. Geng, M. Xu, Z. Wei, and X. Zhou. Learning deep spatiotemporal feature for engagement
 314 recognition of online courses. In *Proceedings of the IEEE Symposium on Computational*
 315 *Intelligence (SSCI)*, pages 442–447, 2019.
- 316 A. Gupta, A. D’Cunha, K. Awasthi, and V. Balasubramanian. Daisee: Towards user
 317 engagement recognition in the wild, 2016. arXiv:1609.01885.
- 318 S. Gupta, T. Ashwin, and R. Guddeti. Students’ affective content analysis in smart class-
 319 room environment using deep learning techniques. *Multimedia Tools and Applications*,
 320 78:25321–25348, 2019.
- 321 S. Gupta, P. Kumar, and R. K. Tekchandani. Facial emotion recognition based real-time
 322 learner engagement detection system in online learning context using deep learning mod-
 323 els. *Multimedia Tools and Applications*, 82(8):11365–11394, 2023.
- 324 K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Pro-*
 325 *ceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*,
 326 pages 770–778, 2016.
- 327 M. He, J. Zhang, S. Shan, M. Kan, and X. Chen. Deformable face net for pose invariant
 328 face recognition. *Pattern Recognition*, 100:107113, 2020.
- 329 S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):
 330 1735–1780, 1997.

- 331 J. Hu, L. Shen, and G. Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE*
332 *Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7132–7141,
333 2018.
- 334 T. Huang, Y. Mei, H. Zhang, S. Liu, and H. Yang. Fine-grained engagement recognition
335 in online learning environment. In *Proc. IEEE 9th Int. Conf. Electron. Inf. Emergency*
336 *Commun. (ICEIEC)*, pages 338–341, 2019.
- 337 E. E. Ketonen, A. Haarala-Muhonen, L. Hirsto, J. J. Hänninen, K. Wähälä, and K. Lonka.
338 Am i in the right place? academic engagement and study success during the first years
339 at university. *Learning and Individual Differences*, 51:141–148, 2016.
- 340 J. Lei Ba, J. R. Kiros, and G. E. Hinton. Layer normalization. *arXiv e-prints, arXiv-1607*,
341 2016.
- 342 J. Liao, Y. Liang, and J. Pan. Deep facial spatiotemporal network for engagement prediction
343 in online learning. *Applied Intelligence*, 51:6609–6621, 2021.
- 344 T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal loss for dense object detection.
345 In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages
346 2980–2988, 2017.
- 347 S. Minaee, E. Azimi, and A. Abdolrashidi. DeepSentiment: Sentiment analysis using en-
348 semble of cnn and bi-lstm models. *arXiv preprint arXiv:1904.04206*, 2019.
- 349 M. Mukhopadhyay, S. Pal, A. Nayyar, P. K. D. Pramanik, N. Dasgupta, and P. Choudhury.
350 Facial emotion detection to assess learner’s state of mind in an online learning system.
351 In *Proceedings of the 2020 5th International Conference on Intelligent Information Tech-*
352 *nology (ICIIT)*, pages 107–115, 2020.
- 353 S. Narduzzi, E. Türetken, J. P. Thiran, and L. A. Dunbar. Adaptation of mobilenetv2
354 for face detection on ultra-low power platform. In *Proceedings of the 2022 9th Swiss*
355 *Conference on Data Science (SDS)*, pages 1–6. IEEE, 2022.
- 356 O. M. Nezami, M. Dras, L. Hamey, D. Richards, S. Wan, and C. Paris. Automatic recogni-
357 tion of student engagement using deep learning and facial expression. In *Joint European*
358 *Conference on Machine Learning and Knowledge Discovery in Databases*, pages 273–289.
359 Springer, 2019.
- 360 X. Niu, H. Han, J. Zeng, X. Sun, S. Shan, Y. Huang, S. Yang, and X. Chen. Automatic
361 engagement prediction with gap feature. In *Proc. 20th ACM International Conference*
362 *on Multimodal Interaction (ICMI)*, pages 599–603, 2018.
- 363 E. Okur, N. Alyuz, S. Aslan, U. Genc, C. Tanriover, and A. A. Esme. Behavioral engagement
364 detection of students in the wild. In *Proceedings of the 18th International Conference on*
365 *Artificial Intelligence in Education (AIED)*, pages 250–261, Wuhan, China, 2017.
- 366 Z. Papamitsiou and A. A. Economides. Learning analytics and educational data mining
367 in practice: a systematic literature review of empirical evidence. *Journal of Educational*
368 *Technology & Society*, 17(4):49, 2014.

- 369 A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin,
370 N. Gimelshein, and L. Antiga. Pytorch: An imperative style, high-performance deep
371 learning library. *arXiv preprint arXiv:1912.01703*, 2019.
- 372 M. Pilotti, S. Anderson, P. Hardy, P. Murphy, and P. Vincent. Factors related to cognitive,
373 emotional, and behavioral engagement in the online asynchronous classroom. *International Journal of Teaching and Learning in Higher Education*, 29(1):145–153, 2017.
- 375 C. Qi, J. Zhang, H. Jia, Q. Mao, L. Wang, and H. Song. Deep face clustering using residual
376 graph convolutional network. *Knowledge-Based Systems*, 211:106561, 2021.
- 377 A. Ramamurthy, V. Sathya, M. Rochman, and M. Ghosh. ML-based classification of device
378 environment using wi-fi and cellular signal measurements. *IEEE Access*, 10:29461–29472,
379 2022.
- 380 I. Roll and P. H. Winne. Understanding, evaluating, and supporting self-regulated learning
381 using learning analytics. *Journal of Learning Analytics*, 2(1):7–12, 2015.
- 382 A. Schulc, J. F. Cohn, J. Shen, and M. Pantic. Automatic measurement of visual attention
383 to video content using deep learning. In *Proc. 2019 16th International Conference on*
384 *Machine Vision Applications (MVA)*, pages 1–6. IEEE, 2019.
- 385 T. Selim, I. Elkabani, and M. A. Abdou. Students engagement level detection in online
386 e-learning using hybrid efficientnet-b7 together with tcn, lstm, and bi-lstm. *IEEE Access*,
387 10:99573–99583, 2022.
- 388 F. M. Shiri, T. Perumal, N. Mustapha, and R. Mohamed. A comprehensive overview
389 and comparative analysis on deep learning models: Cnn, rnn, lstm, gru. *arXiv preprint*
390 *arXiv:2305.17473*, 2023.
- 391 F. M. Shiri, E. Ahmadi, M. Rezaee, and T. Perumal. Detection of student engagement in
392 e-learning environments using efficientnetv2-l together with rnn-based models. *Journal*
393 *of Artificial Intelligence*, 6, 2024.
- 394 C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke,
395 and A. Rabinovich. Going deeper with convolutions. In *Proc. IEEE Conf. Comput. Vis.*
396 *Pattern Recognit.*, pages 1–9, 2015.
- 397 D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri. Learning spatiotemporal
398 features with 3d convolutional networks. In *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*,
399 2015.
- 400 K. Verbert, E. Duval, J. Klerkx, S. Govaerts, and J. L. Santos. Learning analytics dashboard
401 applications. *American Behavioral Scientist*, 57(10):1500–1509, 2013.
- 402 J. M. Vytasek, A. Patzak, and P. H. Winne. Analytics for student engagement. In *Machine*
403 *Learning Paradigms: Advances in Learning Analytics*, pages 23–48. 2020.
- 404 S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma. Linformer: Self-attention with linear
405 complexity. *arXiv preprint arXiv:2006.04768*, 2020.

- 406 J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, and J. R. Movellan. The faces of engagement:
407 Automatic recognition of student engagement from facial expressions. *IEEE Transactions*
408 *on Affective Computing*, 5(1):86–98, 2014.
- 409 P. H. Winne, J. M. Vytasek, A. Patzak, M. Rakovic, Z. Marzouk, and A. Pakdaman-
410 Savoji. Designs for learning analytics to support information problem solving. https://doi.org/10.1007/978-3-319-64274-1_11, 2017.
- 412 J. Wu, B. Yang, Y. Wang, and G. Hattori. Advanced multi-instance learning method
413 with multi-features engineering and conservative optimization for engagement intensity
414 prediction. In *Proc. Int. Conf. Multimodal Interact.*, pages 777–783, 2020.
- 415 H. Zhang, X. Xiao, T. Huang, S. Liu, Y. Xia, and J. Li. An novel end-to-end network for
416 automatic student engagement recognition. In *Proceedings of the IEEE 9th International*
417 *Conference on Electronics Information and Emergency Communication (ICEIEC)*, pages
418 342–345, 2019a.
- 419 J. Zhang, X. Yu, X. Lei, and C. Wu. A novel capsnet neural network based on mobilenetv2
420 structure for robot image classification. *Frontiers in Neurorobotics*, 16:1007939, 2022.
- 421 S. Zhang, X. Pan, Y. Cui, X. Zhao, and L. Liu. Learning affective video features for facial
422 expression recognition via hybrid deep learning. *IEEE Access*, 7:32297–32304, 2019b.
- 423 G. Zhao, X. Sun, J. Xu, Z. Zhang, and L. Luo. Muse: Parallel multi-scale attention for
424 sequence to sequence learning. *arXiv preprint arXiv:1911.09483*, 2019.
- 425 B. Zhu, X. Lan, X. Guo, K. Barner, and C. Boncelet. Multi-rate attention based gru model
426 for engagement prediction. In *Proc. Int. Conf. Multimodal Interact.*, pages 841–848, 2020.
- 427 J. Zhu, T. Chen, and J. Cao. Siamese network using adaptive background superposition
428 initialization for real-time object tracking. *IEEE Access*, 7:119454–119464, 2019.